

Manufacturing Quality Comparison Report

F-Test Statistical Analysis

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Executive Summary

Conclusion: REJECT NULL HYPOTHESIS
The two plants have SIGNIFICANTLY different variance levels (quality consistency) at 5.0% significance level.
Plant with Better Consistency: Plant 1 (99.91% lower variance)
P-Value: 0.000000

Statistical Results

Metric	Plant 1	Plant 2
Sample Size (n)	10	10
Mean	20.010000	21.500000
Variance (s²)	0.014333	15.833333
Std. Deviation (s)	0.119722	3.979112

F-Test Results

Parameter	Value
F-Statistic	1104.651163
Degrees of Freedom (numerator)	9
Degrees of Freedom (denominator)	9
P-Value (two-tailed)	0.000000
Significance Level (α)	0.05
Test Conclusion	REJECT NULL HYPOTHESIS

Interpretation

Null Hypothesis (H₀): The variances of Plant 1 and Plant 2 are equal ($\sigma_1^2 = \sigma_2^2$)
Alternative Hypothesis (H₁): The variances are not equal ($\sigma_1^2 \neq \sigma_2^2$)
Test Result: The two plants have SIGNIFICANTLY different variance levels (quality consistency) at $\alpha = 0.05$

significance level.

Plant with Better Consistency: Plant 1 has lower variance, indicating higher consistency in quality measurements (99.91% improvement).